



SCIENCE FOUNDATIONS

Experimental Error

How good is my data?

PART 1

Where Does Error Come From?



Why We Repeat Experiments



Some Trials Are High, Some Are Low

One measurement is a guess. Repeat it, and some trials come out a bit high, others a bit low.

The HIGHS and LOWS cancel each other out when you AVERAGE them — which brings you close to the true value.

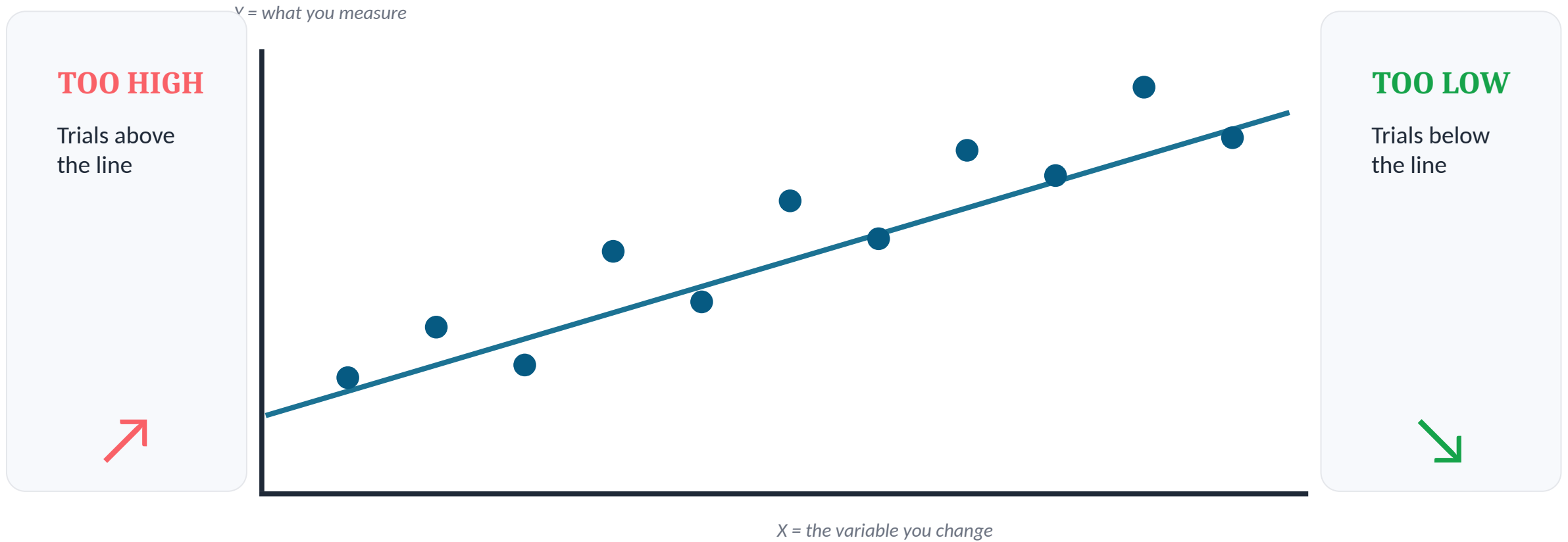


The Trendline IS the Average

When you plot your repeated trials on a graph, the trendline shows the average pattern — it's what you'd expect if random error had cancelled itself out.

More trials → tighter average → better estimate of truth.

Trendline = Average of the Data



The Five Sources of Error

1

YOU

Reaction time, eyesight, technique, mood.

2

Your Tools

Every ruler, scale, or timer has a limit.

3

Manipulation

Nudging or moving the setup while measuring.

4

External influences

A/C, drafts, vibrations, temperature drift.

5

Experimental design

Poor controls or too many variables changing.

PART 2

Precision vs. Accuracy

Two Words That Sound Alike



PRECISION

How close your repeated measurements are to EACH OTHER.

Repeatability.

You can spot precision problems DURING the experiment — if one trial is way off, just repeat it.



ACCURACY

How close your data is to the EXPECTED (true) value.

Hit the target.

You can only calculate accuracy AFTER the experiment — using percent error.

Precision — Spot the Outlier

Time for a runner to finish a race (seconds):



The 27 s is obviously wrong. Repeat that trial — don't average bad data in.

Technique Matters Too

Measuring room length with meter sticks — which is more precise?

✗ Sticks END-to-END IN THE AIR

*Wobble. Shake. Gaps. Bad joints.
Small errors on every seam add up.*

✓ Sticks END-to-END ON THE FLOOR

*Sticks stay flat. Seams are tight.
Repeatable — much more precise.*

PART 3

Error & Percent Error

The Error Formula

$$\text{Error} = | \text{accepted} - \text{experimental} |$$



Why Absolute Value?

The vertical bars | | make the result POSITIVE, whether you measured too high or too low.

Error just tells us HOW FAR OFF — never a negative distance.



Same Units As Original

Error is in the SAME units as your measurement.

Measured 175,000 kg vs. accepted 180,000 kg → Error = 5,000 kg.

Why Isn't Error Enough?

A Blue Whale

Accepted mass: 180,000 kg

You measure: 175,000 kg

Error = 5,000 kg

SOUNDS HUGE!

% Error = 2.8%

A House Cat

Accepted mass: 8 kg

You measure: 7 kg

Error = 1 kg

SOUNDS TINY!

% Error = 12.5%

The Percent Error Formula

$$\% \text{ Error} = (\text{Error} \div \text{Accepted}) \times 100\%$$

*The closer your percent error is to 0,
the more ACCURATE your data.*

THEY ALL MEAN THE SAME THING

You'll See This Formula In Different Ways

MOST TEXTBOOKS

| accepted – experimental |

accepted

× 100%

*Result: no units,
just add a %*

SCIENCE JOURNALS

| observed – true |

true

× 100%

*Result: no units,
just add a %*

LAB REPORTS

| measured – actual |

actual

× 100%

*Result: no units,
just add a %*

Worked Example — Density of Water

You measure water's density as 0.96 g/mL. The accepted value is 1.00 g/mL. Find the error and percent error.

1

IDENTIFY

Accepted = 1.00 g/mL
Experimental = 0.96 g/mL

2

ERROR

$|1.00 - 0.96|$
 $= 0.04 \text{ g/mL}$

3

% ERROR

$(0.04 \div 1.00) \times 100\%$
 $= 4\%$

4

JUDGE

4% error — pretty good
for a middle-school lab.

Practice — Counting Raisins

1

ESTIMATE

How many raisins do you think are in the box?

2

COUNT

Pour them out. Count the actual number.

3

FIND ERROR

Error = |actual - estimate|

4

FIND % ERROR

% Error = (Error ÷ actual) × 100%

HOMWORK

Experimental Error Problem Set #1

Posted on Google Classroom. Due next class meeting.

Remember: identify accepted, experimental, error, and % error — with units.